Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.25	20.25	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.02	+0.0%
power (W), mean	630	641	+1.8% worse
power (W), peak	652	680	+4.1% worse
energy (Wh)	213	216	+1.8% worse
energy, parked workers still on at idle power (Wh)	213	216	+1.8% worse
CPU used (cores), mean	0.921	1.22	+32.8% better
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.038	0.051	+32.8% better
energy per core-hour (Wh)	684	524	-23.4% better
node-hours per core-hour	6.51	4.90	-24.7% better
pending pods, pod-minutes	0.267	0.000	-100.0% better
pending pods, peak	1.00	0.000	-100.0% better
HPA replicas, mean	9.02	9.38	+4.0% worse
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.783	+0.0%
requests timed	1,040	1,100	+5.8%
time over the response line (% of samples)	0.865	0.000	-100.0% better
response time (ms), mean	157	95.40	-39.4% better
response time (ms), median	130	83.40	-35.8% better
response time (ms), 95th percentile	313	154	-50.7% better
response time (ms), 99th percentile	466	196	-57.9% better
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.0639	5.5396	-21.6%	[-3.1234, +0.0177]	no significant difference
kwh_per_core_hour	0.7387	0.5874	-20.5%	[-0.3064, +0.0010]	no significant difference
utilisation	0.0379	0.0501	+32.3%	[+0.0005, +0.0253]	better
pending_pod_minutes_per_hour	0.7619	0.0000	-100.0%	[-2.2857, +0.0000]	no significant difference
hpa_shortfall_minutes_per_hour	2.1964	2.2135	+0.8%	[-3.5587, +3.6805]	no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.359 0.392 0.421 0.446 0.468 0.487 0.504 0.519 0.533 0.545 0.555 0.561 0.566 0.57 0.575 0.58 0.583 0.586 0.588 0.589
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

```
restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
```

Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	618	622	5.00	6.00	0.000	0.000	0.581	0.683

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	619	626	5.00	7.00	0.250	0.000	0.623	0.795
2	6.00	6.00	627	623	6.00	7.00	0.000	0.000	0.833	0.753
3	6.00	6.00	625	635	6.00	8.00	0.000	0.000	0.818	1.05
4	6.00	6.00	635	645	9.00	10.00	0.000	0.000	1.07	1.34
5	6.00	6.00	630	645	10.00	10.00	0.000	0.000	0.956	1.32
6	6.00	6.00	633	642	10.00	10.00	0.000	0.000	1.02	1.26
7	6.00	6.00	644	672	10.00	10.00	0.000	0.000	1.29	2.03
8	6.00	6.00	647	673	10.00	10.00	0.000	0.000	1.37	2.10
9	6.00	6.00	648	676	10.00	10.00	0.000	0.000	1.40	2.13
10	6.00	6.00	649	672	10.00	10.00	0.000	0.000	1.43	2.04
11	6.00	6.00	621	625	10.00	10.00	0.000	0.000	0.649	0.772
12	6.00	6.00	619	622	10.00	10.00	0.000	0.000	0.609	0.745
13	6.00	6.00	624	633	10.00	10.00	0.000	0.000	0.783	1.01
14	6.00	6.00	634	646	10.00	10.00	0.000	0.000	0.995	1.32
15	6.00	6.00	634	648	10.00	10.00	0.000	0.000	1.02	1.38
16	6.00	6.00	633	648	10.00	10.00	0.000	0.000	0.998	1.41
17	6.00	6.00	630	631	10.00	10.00	0.000	0.000	0.908	0.955
18	6.00	6.00	617	628	10.00	10.00	0.000	0.000	0.606	0.881
19	6.00	6.00	619	624	10.00	10.00	0.000	0.000	0.600	0.757
20		6.00		621		10.00		0.000		0.677

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.